Self-cleaning and Non-Scouring Velocity in sewers

A certain minimum velocity has to be maintained to avoid silting and clogging in the sewers. Such a minimum velocity is known as **self-cleaning velocity**.

To keep the sewers free from any trouble, this velocity should be developed at least once in a day, preferably twice.

The maximum permissible velocity at which no scouring action (wear and tear) takes place is known as **non-scouring velocity**.

Non-scouring velocity mainly depends on the materials used in the construction of sewers.

5.4.3 Laying of Sewer lines

In the sewerage system, setting out of alignment is generally started from the tail end on the downstream and proceed upwards. The procedure to be adopted for the laying of a new sewer is as follows

a. Borings (or) trial holes are dug along the proposed sewer line to know the nature of soil.
b. The exact position of manholes are located with the help of sewage plan (or) working drawings.
c. The centre line of sewer between two manholes is marked on the surface of the ground. The length between two successive manholes should be straight.
d. Along the straight line, pegs are driven at convenient distance.

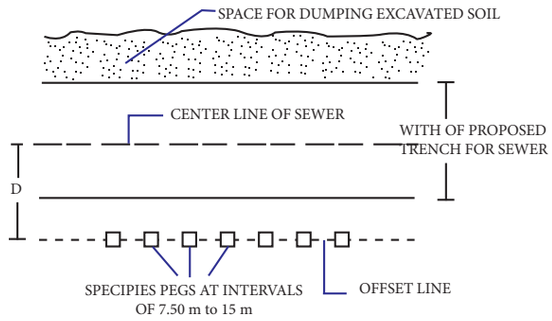
There are two methods for laying the new sewer line. They are:

a. Offset line method
b. Sight rail method

a. Offset line method

In this method, a line parallel to the centre line of the sewer is marked on one side which is known as Offset line.

It is usually marked at a distance of 'D' which is about one half the trench width plus 600mm as shown in fig.

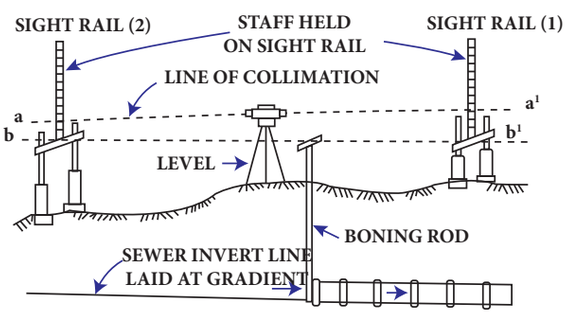


OFFSET LINE

The Offset line helps in locating the centre line of the sewers when excavation is being carried out. The other side is used to dump the material.

b. Sight Rail Method

1. In this method two vertical posts are driven into the ground at a known distance from the centre line of the peg on either side. The horizontal members known as Sight rails are held in position with vertical posts.
2. The line joining the top of the sight rails will give the gradient of the sewer to be laid. This slope is transferred to ground with the help of a boning rod.
3. After laying the sewer pipes, the levels are checked once using a levelling instrument.



Laying of sewer

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